TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

Topic: Customers Management System for NAWACO

Project Code: CMSN

Document Code: CMSN-v1.4

– Hanoi, Jan 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
1/12/2025	A	HoanTX	Add Training Plan, Responsibility Assignment, Project Communication
2/12/2025	A	HoanTX	Add Project Objectives, Project Risk, Project Deliverables, Configuration Managements
10/01/2026	M	DungND	Modify Project Communications, Configuration Management
10/01/2026	A	GiangPT	Add defect rate
11/01/2026	A	HieuDM	Add Responsibility Assignments
17/1/2026	M	HoanTX	Modify document based on new template
27/01/2026	M	GiangPT	Modify defect rate
3/2/2026	M	HoånTX	Update 1.1

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Cost & Time Estimations

#	Work Package	Est. Effort (pds)
1	***Authentication***	
1.1	Sign in (web)	**2**
1.2	Sign in (mobile)	**2**
1.3	Change password (web)	**2**
1.4	Change password (mobile)	**2**
1.5	Forgot password (web)	**2**
1.6	Forgot password (mobile)	**2**
1.7	Control login from unfamiliar devices	**4**
1.8	View profile	**2**
1.9	Update profile	**2**
1.10	System logging	**4**
2.	***Authorization (web)***	
2.1	View website pages	**2**
2.2	Activate/Inactivate website pages	**4**
2.3	View access rights of employees	**4**
2.4	Update access rights of employees	**4**
3	***Roadmap Management (web)***	
3.1	View roadmaps list	**2**
3.2	Update roadmap	**4**
3.3	Delete roadmap	**4**
3.4	Create roadmap	**2**
4	***Lateral Management (web)***	
4.1	View lateral	**2**
4.2	Update lateral	**4**
4.3	Delete lateral	**4**
4.4	Create lateral	**2**

5	***Commune Management (web)***	
5.1	View communes list	**2**
5.2	Update commune	**4**
5.3	Delete commune	**4**
5.4	Create commune	**2**
6	***Neighborhood Unit Management (web)***	
6.1	View neighborhood units list	**2**
6.2	Update neighborhood unit	**4**
6.3	Delete neighborhood unit	**4**
6.4	Create neighborhood unit	**2**
7	***Road Management (web)***	
7.1	View roads list	***2***
7.2	Update road	***4***
7.3	Delete road	***4***
7.4	Create road	***2***
8	***Hamlet (web)***	
8.1	View hamlets list	***2***
8.2	Update hamlet	***4***
8.3	Delete hamlet	***4***
8.4	Create hamlet	***2***
9	***Materials Price (web)***	
9.1	View materials prices list	***2***
9.2	Update materials price	***4***
9.3	Delete materials price	***4***
9.4	Create materials price	***2***
10	***Water Meter Type (web)***	
10.1	View water meter types list	***2***
10.2	Update water meter type	***4***
10.3	Delete water meter type	***4***

10.4	Create water meter type	***2***
11	***Materials Group (web)***	
11.1	View materials groups list	***2***
11.2	Update materials group	***4***
11.3	Delete materials group	***4***
11.4	Create materials group	***2***
12	***Unit (web)***	
12.1	View units list	***2***
12.2	Update unit	***2***
12.3	Delete unit	***2***
12.4	Create unit	***4***
13	***System Parameter Management (web)***	
13.1	View list of system parameters	***2***
13.2	Update system parameters	***2***
14	***Water Price Management (web)***	
14.1	View list of water prices	***2***
14.2	Create new water price	***2***
14.3	Update water price	***2***
14.4	Delete water price	***4***
15	***History of Updating Information Management (web)***	
15.1	View history of updating information	***4***
16	***Department Management (web)***	
16.1	View list of departments	***2***
16.2	Create department	***2***
16.3	Update department	***2***
16.4	Delete department	***4***
17	***Branch (Water Supply Network) Management (web)***	
17.1	View list of branches	***2***
17.2	Create branch	***2***

17.3	Update branch	***2***
17.4	Delete branch	***4***
18	***Employees Management (web)***	
18.1	View list of employees	***2***
18.2	Create employee	***2***
18.3	Update employee	***2***
18.4	Delete employee	***4***
18.5	Lock/Unlock account	***4***
19	***Job Management (web)***	
19.1	View list of jobs	***2***
19.2	Create new job	***2***
19.3	Update job's information	***2***
19.4	Delete job	***4***
20	***New Installation Requests Management (web)***	
20.1	View new installation request list	***2***
20.2	Create new installation request	***4***
20.3	Approve / reject installation requests	***2***
21	***Receipt Management (web)***	
21.1	View receipts list	***2***
21.2	Create receipt	***4***
21.3	Update receipt	***4***
21.4	Delete receipt	***2***
22	***Request solving***	
22.1	Assign requests for processing	***2***
22.2	Create construction pending request	***2***
22.3	Update construction pending request	***2***
23	***Cost Estimates Management (web)***	
23.1	View cost estimates list	***2***
23.2	Create new cost estimate	***4***

23.3	Update cost estimate	***4***
23.4	Approve / reject cost estimaté	***2***
24	***Contract Management (web)***	
24.1	View contracts list	***2***
24.2	Create new contract	***4***
24.3	Delete contract	***4***
25	***Construction Requests Management(web)***	
25.1	View pending construction request	***2***
25.2	Accept construction	***2***
26	***Settlements management (web)***	
26.1	View settlements list	***2***
26.2	Create new settlement	***4***
26.3	Update settlement	***2***
26.4	Sign for confirmation	***2***
27	***Report and Statistics Management (web)***	
27.1	View reports and statistics of requests pending cost estimates	***6***
27.2	View reports and statistics of pending estimate approval requests	***6***
27.3	View reports and statistics of estimation approval–rejected requests	***6***
27.4	View reports and statistics of assigned survey	***6***
28	***Customers Management (web)***	
28.1	View customers list	***2***
28.2	View customer details	***2***
28.3	Update customers information	***2***
28.4	View customer water usage history	***2***
28.5	Delete customer	***4***
28.6	Create customer	***2***
29	***System Portal Management***	
29.1	Get notification (web)	***2***

29.2	Get notification (mobile)	***4***
29.3	Create notification	***4***
30	***Water Meter Recording Management***	
30.1	Capture water meter (mobile)	***4***
31	***This Month's Meter Recordings Management***	
31.1	Verify and validate meter readings (mobile)	***4***
31.2	View customers list of a roadmap	***2***
31.3	View customer's information in the meter recording period	***2***
31.4	View payment information list (mobile)	***2***
31.5	Update index (mobile)	***2***
32	***Arrange meter recording roadmap (web)***	
32.1	Assign meter recording roadmap	***2***
32.2	Cancel meter recording roadmap assignments	***2***
33.3	Update meter recording roadmap assignments	***2***
33.4	View roadmap assignments	***2***
34	***Bills Management (web)***	
34.1	View water consumption volume	***2***
34.2	View bills	***2***

Table 1: Cost & Time Estimations

Total Est. Effort: 360 man-days

1.2 Project Objectives

1.2.1 Allocated Effort

#	Members	Weekdays (hours)	Weekends (hours)
1	Trần Xuân Hoàn	5	8
2	Phạm Hải Đăng	5	8
3	Phạm Trường Giang	5	8
4	Nguyễn Đức Dũng	5	8

5	Dương Minh Hiếu	5	8

Table 2: Allocated Effort

Total allocated effort: 385 man-days

1.2.2 Defect Rate

#	Metric	Unit	Planned	Actual	Notes / References
1	Effort Usage	Person-day	60	52	Need to analyze more for the actual effort distribution (as tracked in the Project Tracking, sheet Scope)
2	Reviewing	%	100	16	Code & design review (Web & Mobile)
3	Unit Test	%	100	10	Unit tests for Web, Mobile, and API
4	Integration Test	%	100	8	Web–API–Mobile integration testing
5	System Test	%	100	5	End-to-end testing across platforms
6	Acceptance Test	%	100	1	User acceptance on Web & Mobile
7	Timeliness	%	100	92	Total on-time deliverables / Total project deliverables
8	Requirement Completeness	%	100	96	Failed if <=75%

Table 3: Defect Rate

1.3 Project Risks

Throughout the development of the Customers Management System NAWACO, several potential risks have been identified that could affect the project's success. These risks encompass various aspects, including team dynamics, technical challenges, project management inefficiencies, data security, and the use of outdated technology. It is crucial to proactively address these risks through appropriate response plans to mitigate their impact and ensure the timely delivery and quality of the project. The table below outlines these risks, their potential impact and likelihood, as well as the response strategies that will be implemented.

#	Risk Description	Impact	Possibility	Response Plans
1	Requirements may arise and	High	Medium	Efforts should be made to work closely and clearly with the client to

	change during development may cause rework and schedule delay			avoid errors and misunderstandings in defining requirements and business processes from the early stages.Freeze requirements after sprint planning. If changes are unavoidable, use the change request process.
2	Deployment issues may occur when releasing to production environment	Medium	Low	Prepare deployment checklist, staging environment testing
3	Data inconsistency may occur during data migration or synchronization	High	Low	Validate data migration scripts, backup before migration
4	Critical defects may be discovered late in system testing	High	Low	Prioritize core use cases, regression testing, early system testing
5	Initial task effort estimation may be inaccurate, causing schedule slippage during implementation	High	Medium	Break tasks into smaller units, review estimates after each sprint
6	Fixing defects may introduce additional changes, increasing workload beyond plan	Medium	Medium	Apply regression testing, limit scope of fixes
7	A team member may become temporarily unavailable due to personal reasons, delaying assigned tasks	Medium	Low	Cross-train team members, reassign tasks, maintain task documentation

Table 4: Project Risks

2. Management Approach

The Customers Management System for NAWACO is developed using an Iterative and Incremental Development Model, ensuring continuous enhancement through multiple cycles of refinement and systematic expansion. Each iteration produces a functional version of the system, allowing for continuous testing, feedback, and improvements. This approach reduces development risks, adapts to changing requirements, and ensures that core functionalities, the system archives better performance, user satisfaction, and long-term sustainability.

2.1 Project Processes

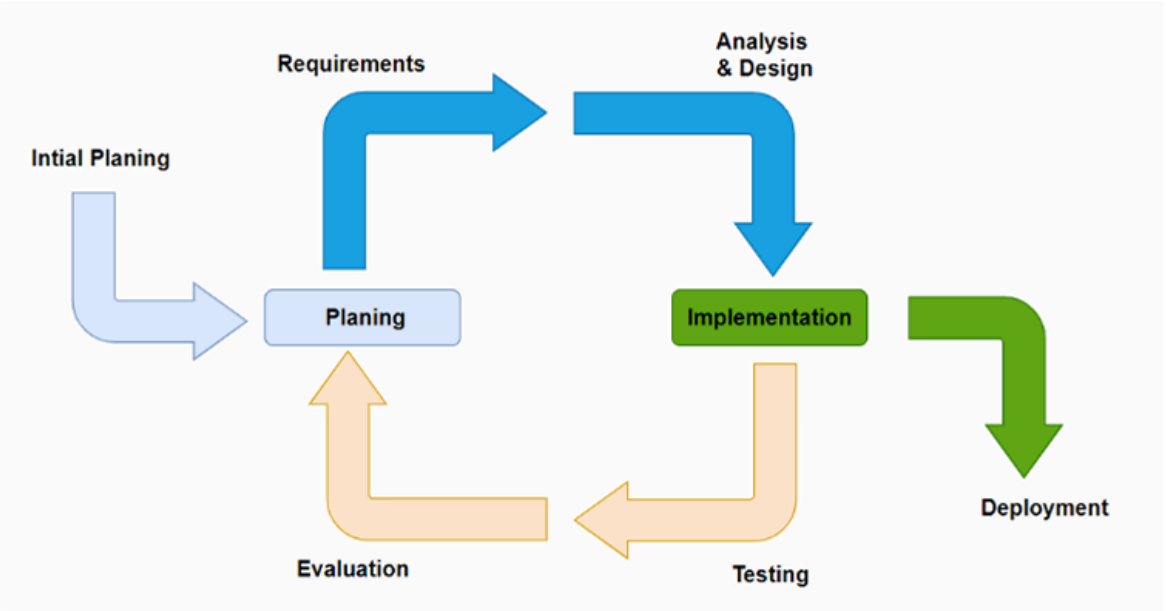


Figure 1: Iterative and Incremental Model Process

The software development process model applied to the CMSN project follows the Iterative and Incremental Development Model. This approach ensures continuous improvement through multiple cycles of refinement and systematic expansion.
In this project, we plan to divide it into 4 specific iterations as follows:

- **Iteration 1 (3 weeks - from 5/1/2026 to 25/1/2026)**
  - Conduct the kick-of Meeting
  - Survey the main business and workflow
  - Define the Project Scope and Features
  - Prepare Project Introduction (Report 1 - Version 1.0)
  - Define functions for Iteration 1 and develop the Project Plan (Report 2 - Version 1.0)
  - Establish high-level requirements, analyze detailed requirements for Iteration 1, and initiate the Software Requirement Specification (Report 3 - Version 1.0)
  - Develop high-level design, analyze detailed design for iteration 1 features, and begin the Software Design (Report 4 - Version 1.0)

- Coding the planned functions
- Conduct testing and bug fixing (Report 5.1, 5.2, 5.3 - Version 1.0)
- Assess the effectiveness of the iteration

- **Iteration 2 (3 weeks - from 26/1/2026 to 15/2/2026)**
  - Define functions for Iteration 2 and revise the Project Plan (Report 2 - Version 1.1)
  - Revise requirements (if requirements change); analyze detailed requirements for iteration 2 features; update design for iteration 2 features and update Software Requirement Specification (Report 3 - Version 1.1)
  - Analyze detailed design for iteration 2 features and update Software Design Document (Report 4 - Version 1.1)
  - Proceed with coding the planned functions
  - Conduct testing and bug fixing (Update report 5.1, 5.2, 5.3 - Version 1.1)
  - Evaluate iteration effectiveness

- **Iteration 3 (2 weeks - from 16/2/2026 to 15/3/2026)**
  - Define functions for iteration 3 and revise the Project Plan (Report 2 - Version 1.2)
  - Revise requirements (if requirements change) and analyze detailed requirements for iteration 3 features and update Software Requirement Specification (Report 3 - Version 1.2)
  - Analyze detailed design for iteration 3 features and update Software Design Document (Report 4 - Version 1.2)
  - Proceed with coding the planned functions
  - Conduct testing and bug fixing (Update report 5.1, 5.2, 5.3
  - Version 1.2)
  - Evaluate iteration effectiveness

- **Iteration 4 (3 weeks - from 16/3/2026 to 5/4/2026)**
  - Define functions for iteration 4 and revise the Project Plan (Report 2 - Version 1.3)
  - Revise requirements (if requirements change) and analyze detailed requirements for iteration 4 features and update Software Requirement Specification (Report 3 - Version 1.3)
  - Analyze detailed design for iteration 4 features and update Software Design Document (Report 4 - Version 1.3)
  - Proceed with coding the planned functions
  - Conduct testing and bug fixing (Update report 5.1, 5.2, 5.3 - Version 1.3)
  - Proceed with project deployment
  - Evaluate iteration effectiveness
  - Proceed with project deployment
  - Prepare User Guide (Report 6)
  - Evaluate iteration effectiveness
  - Final Report (Report 7)
  - Submit The Capstone Project

- Prepare Thesis Defense Slides and Practical Presentation
- Evaluate project effectiveness

2.2 Quality Management

To ensure the CMSN system is reliable, secure, and efficient, we implement a structured quality management approach. This includes defect prevention, reviewing, and multiple levels of testing to meet project quality objectives.

2.2.1 Defect Prevention

Defect prevention is planned to minimize errors at early stages of development. The project emphasizes requirement clarity, standardized coding practices, and early design validation to reduce defect introduction.

Based on the estimated defect distribution, a higher defect rate is expected during the reviewing phase, as early reviews aim to identify potential issues in requirements, design, and source code before implementation and testing. Early defect detection helps reduce risks and rework costs in later stages.

2.2.2 Reviewing

Reviewing is planned as the primary early quality control activity and is conducted throughout all iterations (Iteration 1 to Iteration 4). This activity includes requirement reviews, design reviews, and code reviews for Web, Mobile, and API components.

Reviews are performed at the beginning of each iteration before implementation and during development to ensure requirement clarity, design consistency, and code quality. The reviewing stage is planned with 100% coverage and is expected to detect the majority of defects before entering testing phases.

Reviewing methods include peer reviews and supervisor reviews, supported by document checklists and coding standards. Tools such as Microsoft Word, Google Docs, and GitHub Pull Requests are used to manage review feedback and version control.

2.2.3 Unit Testing

Unit testing is planned for Iterations 1 to 4 and is conducted immediately after coding individual modules. Developers are responsible for executing unit tests to verify the correctness of Web, Mobile, and API components.

Unit testing focuses on validating individual functions in isolation before integration. Automated and manual testing approaches are applied depending on module complexity. Testing tools include framework-specific testing libraries and Postman for API validation.

2.2.4 Integration Testing

Integration testing is planned from Iteration 1 onward and becomes more extensive in Iterations 2 and 3 as more system components are integrated. This testing phase verifies interactions between modules, particularly Web–API–Mobile communication.

Integration testing is conducted after unit testing within each iteration. Test cases focus on data flow, interface compatibility, and workflow correctness. Tools such as Postman and browser-based testing tools are used to validate integrated functionalities.

2.2.5 System Testing

System testing is planned mainly for Iterations 3 and 4, when major system functionalities are completed. This testing validates the complete CMSN system against functional and non-functional requirements.

System testing includes end-to-end scenarios across Web and Mobile platforms, focusing on performance, stability, and usability. Manual testing and checklist-based validation are applied to ensure full system coverage.

2.2.6 Acceptance Testing

Acceptance testing is planned exclusively for Iteration 4, prior to system deployment. This phase ensures that the system meets user expectations and business requirements.

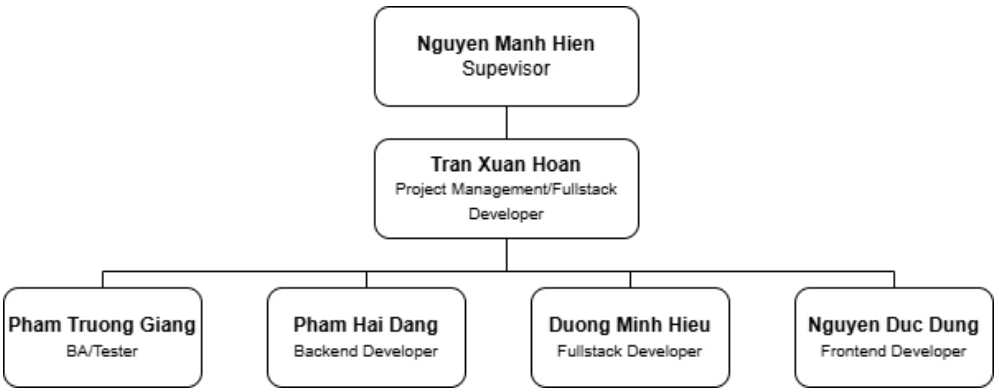
Acceptance testing is conducted based on predefined acceptance criteria and real usage scenarios. Feedback from acceptance testing is documented and addressed before final deployment. Tools include test case documents and user feedback forms.

2.3 Project Training Plan

Training Area	Participants	When, Duration
Microservice architecture	All team	1/1/2026, one day
Spring boot	DangPH	17/1/2026, one day
Next.js	HoanTX, DungND, HieuDM	18/1/2026, one day
Git, Github	All team	1/1/2026, one day
Nx tool	All team (except of GiangPT)	18/1/2026, one day
AI	DangPH	20/1/2026, one day

Table 5: Project Training Plan

3. Responsibility Assignments



RACI Chart: R~Responsible, A~Accountable, C~Consulted, I~Informed

Work Package	HoanTX	DangPH	HieuDM	GiangPT	DungND
Document and Report					
Project Tracking	R	I	I	I	I
Prepare Project Introduction Document	R	R	R	R	R
Prepare Project Management Plan	R	R	R	R	R
Prepare SRS Document	C	R	C	R	R
Prepare SDS Document	R	R	R	R	R
Prepare Test Documentation	I	I	I	R	I
Prepare Software User Guide	R	R	R	R	R
Prepare Final Project Report	R	R	R	R	R
Prepare Project Schedule	R	I	I	I	I

Prepare Presentation Slide	R	R	R	R	R
Design					
System Architecture Design	R	I	C	I	R
Class Diagram	C	C	R	R	I
Sequence Diagram	C	C	R	R	R
Use Case Specification	C	R	C	R	R
Screen Design	R	R	R	R	R
Database Design	R	R	C	I	I
Implementation					
Control source code	R	I	I	I	R
Code Front-end	R	I	R	I	R
Code Back-end	R	R	R	I	I
Code mobile	R	I	R	I	R
Deploy	C	R	C	I	R
Test					
Create test plan	I	I	I	R	I
Create test report	I	I	I	R	I

Table 6: Responsibility Assignments

4. Project Communications

Effective communication is one of the essential elements to ensure the successful completion of the Customers Management System for NAWACO project. The team will hold regular meetings to ensure that all stakeholders are aligned with the project's goals and timeline. These meetings will keep the team informed about project progress, address any issues that arise, and clarify any ambiguities with the client and supervisor. The following table outlines the key communication topics, including the participants, purpose, frequency, and communication methods for each meeting.

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Weekly meeting	All members	Tracking progress of teammates, finding their difficulties and answer necessary questions	8:30 PM, Tuesday of every week	Online or Offline
Iteration meeting	All members	Summarizing the last sprint's work and implement work of current sprint	8:30 PM, Monday for each 3 weeks	Online meetings
Weekly meeting with supervisor	All members	Reviewing all reports, receiving advises, discuss and solve problems	One day / week	Offline
Meeting with customer	All members, Customer	Receive requirements and confirm requirements and ask the customer questions that the team has not yet clarified	Once every 2 months	Offline
Emergency meeting	All members	Discuss and find the best solution in priority order	When there is a problem that needs to be dealt with immediately	Online meetings

Table 7: Project Communications

5. Configuration Management

5.1 Document Management

5.1.1 Centralize Storage

- All documents will be stored in a shared **Google Drive** folder with accessibility for all team members.
- After each increment, documents will be updated in the source code (host by Github)

5.1.2 Version Control

- Documents will follow a clear versioning convention.
  E.g: Report1_Project Introduction_v1.0, Report3_SRS_v1.0, Report4_SRS_v1.1
- After each reviewing phase or major update, the version will be incremented:
  - **Major (X.0):** Significant structural or content changes
  - **Minor (0.x):** Small revisions, corrections, or updates

5.1.3 Review & Approval

- Documents will be reviewed by designated members as assigned in the Responsibility Assignment Matrix
- Once reviewed, documents are marked as "Approved" in the change log

5.1.4 Backup & Recovery

- A weekly backup of the entire document repository will be made to an external storage drive/cloud
- In case of accidental deletion or corruption, files can be recovered from previous backups or version history on Google Drive

Category	Descriptions
Management tools	Google Drive
Editors	Word, Excel, Canva, Figma, Balsamiq
Diagramming	DrawIO, Visual Paradigm
Naming rules	ReportNumber_Document name_version Ex: Report1_ProjectManagementPlan_v1.0
Rules	Each report will be uploaded and tracked by version, responsible person, and upload date. Documents must be reviewed by other members.

Table 8: Document Management

5.2 Source Code Management

Source code management is a crucial aspect of the software development process, as it helps maintain the stability and quality of the system. In this project, GitHub is used as the source code management platform, and strict guidelines are followed to ensure that the development process is carried out efficiently and in an organized manner.

5.2.1 Repository Hosting

- We use **Git** as the version control system
- **Github** is selected as the central remote repository hosting platform for both frontend and backend source code
- All team members are added as collaborators with appropriate access rights

5.2.2 Code Review and Pull Requests

- Teammates implement their features by creating new branches with prefix feat, refactor, style or fix depending on the branch's purpose.
- A **Pull Request (PR)** must be created when teammates want to merge their code into the dev branch.
- After each incremental, the dev branch will be merged into the main branch.
- At least one peer review is required before merging. The participant that is responsible for reviewing is the HoanTXHE account for BE repository and NguyenDucDungHE171774 for FE repository.

5.2.3 Github Workflow

- Clone and pull the repository
- Switch or create branches
- Commit and push changes
- Open pull request

5.2.4 Backup & History

- Git inherently maintains commit history, which allows rollback
- Github serves as an off-site backup and supports change tracking and revert options

Category	Description
Project name	Source code: Capstone Project
Management tools	Github
Editors	Visual Studio Code, IntelliJ
Accessibility	All members
Rules	Code must be reviewed before merge.

	All coding branches must be based on main.
	The branch name convention: <commit type>/<name>-...-<name> ex: feat/create-common-components
	Commit convention: <commit type>: <message>
	Member needs to request merge with leader (HoanTXHE or NguyenDucDungHE171774) to merge code into main or create pull request

Table 9: Version control and rules

5.3 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	Next.js (FrontEnd), React Native (Mobile App), Microservice (Java Spring boot, Node.js) (BackEnd)
Database	PostgreSQL, MongoDB
IDEs/Editors	Visual Studio Code, IntelliJ
Diagramming	Visual Paradigm, DrawIO
Documentation	Microsoft Office, Google Docs/Sheets/Slides
Version Control	Github (Source Codes), Google Drive (Documents)
Deployment server	Vercel, VPS
Project management	Trello (Schedule, Bug-Defect, Tasks), Github (Issues)

Table 10: Tools and Infrastructures